Shubhadeep Mandal

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Education

International Institute of Information Technology, Hyderabad — July 2025 – Present
Telangana, India
Master of Technology in Computer Science and Engineering — *CGPA: 9.12*

Indian Institute of Information Technology, Ranchi — Aug 2020 – Jun 2024
Ranchi, India
Bachelor of Technology (Hons.) in Computer Science and Engineering — *CGPA: 9.05*

Projects

Warehouse Traffic Control: Distributed Robot Coordination | *JavaScript, Three.js, Web Workers*

- Designed a lease-and-fencing protocol coordinating 200 robots across 16 region managers, sustaining **0 collisions over 416K moves** under 40% message loss, ±20% clock drift and injected crashes.
- Implemented Chandy–Misra–Haas deadlock detection and crash recovery that rebuilds a manager's state in ∼4.3 s.
- Built non-blocking gossip congestion control on lease replies, raising tasks completed by **98%** at 150 robots.
- Versus one central coordinator: **7.6x lower peak per-node load** (125 vs 950 msgs/s), and a region-manager crash cuts off **only 11 robots instead of all 120**.

Negotiate, Then Solve: LLM Agents for Timetabling | *Python, Gemini, Qwen, RAG, CP-SAT, FastAPI*

- Built an agentic workflow that turns free-text requests from email, chat and a web portal into typed constraints, resolving conflicts via CP-SAT analysis and LLM negotiation through typed tool calls.
- Designed guardrails (authority checks, cited rules, human approval) so that **no attack succeeded** across 50 prompt-injection, spoofing and data-leak tests, even with a fully compromised parser.
- Built hybrid BM25 + dense-embedding semantic retrieval over OCR-ingested PDFs, scans and web pages, raising policy Recall@5 to **93.4%** (vs 87.7% BM25), with every denial required to cite a retrieved rule.
- Distilled 2.7K multi-task examples from every agent call and LoRA fine-tuned Qwen3.5-4B into a quantised GGUF served by llama.cpp on a 6 GB laptop GPU, with automatic rule-based fallback.

IndicLLM: Transformer Pretraining & Reasoning SFT | *PyTorch, FlashAttention-2, RoPE*

- Pretrained 25M-param decoder-only Transformers from scratch on 1B+ tokens across Hindi and Assamese using bare PyTorch primitives, driving a +36.5% perplexity reduction via FlashAttention-2, RoPE, and SwiGLU.
- Trained dedicated 16k SentencePiece BPE tokenizers on AI4Bharat Sangraha, eliminating token fertility inflation and achieving 0.505 Bits-Per-Byte sequence compression across both high and low-resource languages.
- Fine-tuned 8 model variants across 20k synthetic multi-hop reasoning tasks, unlocking a +45.8% relative Token F1 gain and 61.3% decomposed CoT step score over direct SFT via intermediate deductive scratchpads.

Peer-to-Peer Distributed File Sharing System | *C++, POSIX Sockets, Multithreading, OpenSSL, Linux*

- Built a BitTorrent-style P2P system in C++; self-scheduling workers cut 1GB single-peer downloads from **193s to under 13s**.
- Implemented out-of-order parallel chunk writing directly to designated file offsets and dynamic bitfield generation for real-time partial seeding during active downloads.
- Architected Primary-Backup tracker failover with race-free election and state sync; **46 unit tests** pass under ASan/UBSan.

Technical Skills

Languages: Python, C/C++, JavaScript, TypeScript, Rust, SQL, Bash.

Systems & Distributed: Leases & Fencing, Deadlock Detection, POSIX Sockets, Multithreading, Linux, ASan/UBSan.

AI & ML: LLM Agents, Tool Calling, RAG, LoRA, Pretraining & SFT, PyTorch, FlashAttention-2, llama.cpp.

Web & Tools: FastAPI, React, Three.js, Web Workers, CP-SAT, pytest, Git, GitHub Actions, Docker.

Achievements

- **B.Tech Rank Holder:** Ranked 2nd in IIIT Ranchi by GPA in B.Tech CSE.
- **PGEE 2025:** Secured All-India Rank 100 in IIIT-Hyderabad Post Graduate Entrance Examination (PGEE).
- **Megathon 2025:** Third rank team in Megathon 2025 in the mobile computing track.

Interests

- Web Novel and Manga Enthusiast.
- Chess and Badminton.